

Bank Account Simulation

Description:

UTC Leeds Bank is a new bank to the industry, they have hired you to build them their first banking system called “UTC Leeds Banking App”.

You need to design and build some python code which will link to an sqlite3 database to allow users to sign up for an account with UTC Leeds Bank.

At registration, a new customer (user) will need to give an email address (as a login name), a password (meeting typical rules), First name, surname, Phone number and an opening balance for their account.

Once this has been completed, the system will generate them a USER ID and an Account Number (auto incrementing integer). And log the opening balance as a transaction.

The System should allow a registered user to log in and manage their account including making a deposit to add to their balance, make a withdrawal to remove money from their account, make a purchase and check their remaining balance.

Each transaction should be recorded in the database and registered users should be able to view their transaction history as well as giving them a total amount they have deposited (including the opening balance), the total they have spent on purchases and the total withdrawn.

Task 1: IN GROUPS

Develop an ER diagram to show the basic structure of your 3 table relational database.

Develop that ER diagram to show the fields in each table with appropriate primary and foreign keys.

Design the key algorithms for your software e.g. creating a new user, login system, deposit, withdraw, purchase, transaction history. These should be presented as Flowcharts and / or Pseudocode.

TIP: These designs should be detailed enough that anyone in another group could take them away and mostly build the same system as you intended.

Task 2: INDIVIDUALLY

Build your sqlite3 database using DB Browser app. Save it to the same folder as your Python file will be in.

Task 3: INDIVIDUALLY

Build you banking app, by connecting Python code to the database and implementing all of the features you have been asked to create.

Testing should be completed and evidence as you have gone along. With evidence of how failed tests were resolved.